Jason St George

Principal Systems Architect — AI, Autonomy & High-Constraint Infrastructure

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U.S. Citizen | Active TS/SCI

Summary

Systems architect, technical founder, and ML/software engineer with 7+ years building intelligent and distributed systems under difficult operational constraints. I specialize in ambiguous problems where architecture matters: real-time sensor AI, air-gapped ML infrastructure, adversarial distributed systems, agent and retrieval infrastructure, and 0→1 product platforms. I work from system definition through critical implementation, deployment, and operational handoff.

Signature Outcomes

- Led a DSP+ML underwater detection pipeline achieving **sub-50 ms Jetson inference**, sustained end-to-end on embedded hardware under operational noise.
- Built distributed LLM fine-tuning and serving for air-gapped classified environments across a **multi-GPU cluster**; reduced fine-tuning costs approximately **65%** through quantized adapter-based methods.
- Built and led production of a **ciphertext-only adversarial storage network** with ongoing integrity proofs, anti-precompute challenges, and tiered incentives, operated in production under continuous verification.
- Co-founded Alchemical AI and built **TurnkeyHQ**, a multi-tenant vertical AI operating system, owning product architecture, application and data layers, integrations, release infrastructure, tenant deployment, QA, and lifecycle management.

Work Experience

Co-Founder & Principal Engineer — 2024–Present

Alchemical AI LLC | TurnkeyHQ case study | alchemicalai.com

- Conceived and built TurnkeyHQ for real-estate professionals, carrying the product from market thesis and domain model through production deployment, customer configuration, maintenance, and commercial iteration.
- Owned the complete technical surface: multi-tenant application architecture, agent workflows, data model, auth, third-party integrations, observability, migrations, infrastructure, release engineering, QA, and incident response.
- Developed repeatable tenant onboarding and deployment across a 14-service topology and nine deployable images; created controls for build provenance, environment validation, smoke testing, rollback readiness, and post-deployment verification.
- Managed roadmap tradeoffs and the full product lifecycle while serving as the engineering, platform, release, and QA function.

Senior Machine Learning Software Engineer — July 2020–Present

Black River Systems Co., Utica, NY

- **Owned** distributed LLM fine-tuning capability in a TS/SCI SCIF, from data pipelines and training orchestration through deployment and operator workflows.
- Shipped online learning for UAV fleets with bandwidth-aware synchronization; reduced data egress **60%** while maintaining model quality.
- Delivered Jetson-deployed sonar target recognition combining DSP feature extraction, DeepCNN inference, and end-to-end latency measurement for real-time operator workflows.

Work Experience (continued)

Data Scientist I June 2019–May 2020
Northeast Information Discovery, Inc., Canastota, NY

- Improved RF modulation classification accuracy 12pp using attention-based spectral encoders; built APIs and deployed models into streaming production environments with R, Python, and C++.

Selected Systems & Research

Agentic Data | jasonstgeorge.com/agentic-data 2025

- Architected an enterprise context graph and retrieval planner: 27 endpoints, 21 Postgres tables, 35 edge types, five retrieval strategies, multi-hop expansion, institutional connectors, and 315 tests.

SwarmOS | jasonstgeorge.com/swarmos 2025

- Built an auditable multi-agent research substrate: 13-package TypeScript monorepo, 30+ authenticated APIs, event-stream coordination, content-addressed artifacts, structured claim graphs, verification grades, and budgeted execution.

AfterFiat — Sole Author, v3.0 | Research profile 2024–Present

- Wrote and maintains a **checksummed, versioned monetary research thesis** testing whether Privacy, Proofs, and Compute can support store-of-value premium under sustained repression: ten premises, a nine-link chain, eighteen red lines, and four telemetry families.
- Separates utility from moneyness, protocol value capture from market price, and duration-neutral base assets from duration-bearing infrastructure credit; publishes section-level citations and formal release notes.

Published Research

Google Scholar

Music Style Transformer — St. George, J., Bischof (2019). Bayesian-factorized raw-audio transcription, composition, and synthesis; transformer-based generation. *ICAI '19*, pp. 22–33.

Sonification of Black Hole Merger Data — St. George, J., Bischof, H.P., Kim, S.Y. (2018). Velocity-modulated synthesis and HRTF spatialization of CCRG simulations. *MSV '18*, pp. 3–9.

Technical Capabilities

- **Languages / AI:** Python, TypeScript/JavaScript, Rust, SQL, C++; PyTorch, LoRA/QLoRA/PEFT, edge inference, DSP+ML, retrieval planning, agent orchestration, evaluation
- **Platform / Data:** FastAPI, PostgreSQL/pgvector, Neo4j, OpenSearch, Redis Streams, MinIO/S3, Docker, Kubernetes, Slurm, JWT/RBAC, CI/CD, observability
- **Optimization / Protocols:** OR-Tools CBC, MILP, Pedersen commitments, Merkle proofs, challenge-response verification, incentive design, threat modeling

Education

M.S., Computer Science — Rochester Institute of Technology May 2019

B.M., Performance; Music Theory Minor — University of North Texas May 2013